

Eigenvalues and Principal Component Analysis (PCA)

The Foundation of Dimensionality Reduction

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① Eigenvalues & Eigenvectors (The Basics)

- Definition and Geometric Meaning
- Numerical Example: 2×2 Calculation

② Principal Component Analysis (PCA)

- Motivation: The Curse of Dimensionality
- Step-by-Step Algorithm
- Numerical Example: Data Transformation

Motivation: Why Eigenvalues?

- **Linear Transformation:** When a matrix A acts on a vector $\mathbf{v}$, it typically changes both the magnitude and the direction of $\mathbf{v}$.
- **The Special Case:** We are looking for special vectors $\mathbf{v}$ that, when multiplied by A , only change in **magnitude**, not direction.
- This special direction is the **Eigenvector** ($\mathbf{v}$).
- The scaling factor is the **Eigenvalue** (λ).

The Eigenvalue Equation:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Deriving the Characteristic Equation

- 1 Start with the core equation:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

- 2 Rearrange the terms: (Note: $\mathbf{I}$ is the Identity Matrix)

$$\mathbf{A}\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}$$

- 3 Factor out the vector $\mathbf{v}$:

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$$

- 4 **The Condition:** For a non-trivial solution (i.e., $\mathbf{v} \neq \mathbf{0}$), the matrix $(\mathbf{A} - \lambda\mathbf{I})$ must be non-invertible.

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

This is the **Characteristic Equation**.

Case 1: If $\mathbf{B}$ is Invertible

- **Assumption:** Suppose the matrix $\mathbf{B} = (\mathbf{A} - \lambda\mathbf{I})$ is **invertible** (i.e., $\mathbf{B}^{-1}$ exists).
- **The Result:** We multiply the system by $\mathbf{B}^{-1}$:

$$\mathbf{B}^{-1}(\mathbf{B}\mathbf{v}) = \mathbf{B}^{-1}\mathbf{0}$$

$$\mathbf{I}\mathbf{v} = \mathbf{0}$$

$$\mathbf{v} = \mathbf{0}$$

- **Conclusion:** If $\mathbf{B}$ is invertible, the system **only** has the **trivial solution** ($\mathbf{v} = \mathbf{0}$). This is useless for finding eigenvectors.

Case 2: The Necessary Condition

Condition for Non-Trivial Solutions

To guarantee the existence of a vector $\mathbf{v} \neq \mathbf{0}$ that satisfies $\mathbf{B}\mathbf{v} = \mathbf{0}$, the matrix $\mathbf{B}$ **must fail to be invertible**.

- A non-invertible matrix has a non-trivial **Null Space** (Kernel).
- This Null Space contains all the desired non-zero eigenvectors.

The Characteristic Equation

This condition of non-invertibility is mathematically enforced using the determinant:

$$\mathbf{B} \text{ is non-invertible} \iff \det(\mathbf{B}) = 0$$

Substituting $\mathbf{B}$ back in, we get the required **Characteristic Equation**:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

Example: Calculating Eigenvalues (Part 1/5)

Setup and Characteristic Equation

- Let's use a simple 2×2 matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$$

- Construct the required matrix $\mathbf{A} - \lambda\mathbf{I}$:

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 - \lambda & 1 \\ 2 & 2 - \lambda \end{pmatrix}$$

Example: Calculating Eigenvalues (Part 2/5)

Solving $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$

- 1 Set the determinant to zero:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (3 - \lambda)(2 - \lambda) - (1)(2) = 0$$

- 2 Expand the equation:

$$6 - 3\lambda - 2\lambda + \lambda^2 - 2 = 0$$

- 3 Simplify into the characteristic polynomial:

$$\lambda^2 - 5\lambda + 4 = 0$$

Example: Calculating Eigenvalues (Part 3/5)

Finding the Eigenvalues (λ_1 and λ_2)

- We solve the quadratic equation:

$$\lambda^2 - 5\lambda + 4 = 0$$

- Factor the polynomial:

$$(\lambda - 4)(\lambda - 1) = 0$$

- The two Eigenvalues are:

$$\lambda_1 = 4 \quad \text{and} \quad \lambda_2 = 1$$

Example: Calculating Eigenvectors (Part 4/5)

Solving for $\mathbf{v}_1$ when $\lambda_1 = 4$

- ① Substitute $\lambda_1 = 4$ back into $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 3-4 & 1 \\ 2 & 2-4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- ② This simplifies to the system of equations:

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- ③ Both equations are equivalent ($-x_1 + x_2 = 0$ and $2x_1 - 2x_2 = 0$).

$$\mathbf{x}_1 = \mathbf{x}_2$$

Example: Calculating Eigenvectors (Part 5/5)

Result for $\lambda_1 = 4$

- Since $x_1 = x_2$, we choose a simple, non-zero vector, e.g., $x_1 = 1, x_2 = 1$.
- **Eigenvector 1:**

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- **Check:** $\mathbf{A}\mathbf{v}_1 = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$
- And $\lambda_1 \mathbf{v}_1 = 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$. ✓

Finding $\mathbf{v}_2$ when $\lambda_2 = 1$

The Second Eigenvector

- ① Substitute $\lambda_2 = 1$ back into $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 3-1 & 1 \\ 2 & 2-1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- ② This simplifies to the system:

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- ③ The equation is $2x_1 + x_2 = 0$. So, $x_2 = -2x_1$.

Result for $\lambda_2 = 1$

- Let $x_1 = 1$. Then $x_2 = -2$.
- **Eigenvector 2:**

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

- **Check:** $\mathbf{A}\mathbf{v}_2 = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$
- And $\lambda_2 \mathbf{v}_2 = 1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$. ✓

Eigen-Summary and Diagonalization

Summary of Results

Eigenvalue (λ)	Eigenvector ($\mathbf{v}$)
$\lambda_1 = 4$	$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
$\lambda_2 = 1$	$\mathbf{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

- **Geometric Meaning:** These vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are the fundamental directions of the transformation A .

Introduction to Principal Component Analysis (PCA)

- **The Goal: Dimensionality Reduction** while retaining as much information (variance) as possible.
- **The Problem:** High-dimensional data is difficult to visualize, slow to process, and often suffers from the "Curse of Dimensionality" (sparse data).
- **PCA's Solution:** Find a new set of orthogonal axes (the **Principal Components**) that capture the maximum variance in the data.

The PCA Algorithm: 5 Main Steps

- 1 **Standardize** the Data.
- 2 Calculate the **Covariance Matrix**.
- 3 Compute the **Eigenvalues** and **Eigenvectors** of the Covariance Matrix.

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- 1 **Standardize** the Data.
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- 4 Select the top k Eigenvectors (Principal Components).
- 5 **Project** the data onto the new subspace.

PCA Step 1: Standardization

Centering the Data

- 1 **Calculate the Mean:** Find the mean (μ) for each feature (column).
- 2 **Center the Data:** Subtract the mean from every data point in its respective column.

$$x_{centered} = x - \mu$$

- 3 **Why?:** PCA is sensitive to scale. Centering ensures the first Principal Component (PC) is not simply the feature with the highest magnitude. The mean of the centered data is now 0.

PCA Step 2: The Covariance Matrix

Measuring Variance and Correlation

- The Covariance Matrix (**C**) is a square matrix that summarizes the statistical relationship between the features.
- **Diagonal Elements:** Variance of each feature ($Cov(X, X)$).
- **Off-Diagonal Elements:** Covariance between features ($Cov(X, Y)$).

$$\mathbf{C} = \begin{pmatrix} \sigma_X^2 & Cov(X, Y) \\ Cov(Y, X) & \sigma_Y^2 \end{pmatrix}$$

- Note: $Cov(X, Y) = Cov(Y, X)$, so the covariance matrix is always **symmetric**.

PCA Example: Data Setup

Working with a $2D$ Centered Dataset

- **Data Set ($\mathbf{X}$):** Assume the data has already been centered ($\mu = 0$ for both features).

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \\ 2 & 0 \\ -1 & -1 \\ -2 & -2 \end{pmatrix}$$

- **Calculation Setup:** $\mathbf{C} = \frac{1}{N-1} \mathbf{X}^T \mathbf{X}$ (where $N = 5$)

$$\mathbf{X}^T \mathbf{X} = \begin{pmatrix} 1 & 0 & 2 & -1 & -2 \\ 1 & 2 & 0 & -1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 2 \\ 2 & 0 \\ -1 & -1 \\ -2 & -2 \end{pmatrix} = \begin{pmatrix} 1+0+4+1+4 & 1+0+0+1+4 \\ 1+0+0+1+4 & 1+4+0+1+4 \end{pmatrix}$$

PCA Example: Covariance Matrix

Step 2 Result

① Sum of Products:

$$\mathbf{X}^T \mathbf{X} = \begin{pmatrix} 10 & 6 \\ 6 & 10 \end{pmatrix}$$

② The Covariance Matrix (C):

$$\mathbf{C} = \frac{1}{5-1} \begin{pmatrix} 10 & 6 \\ 6 & 10 \end{pmatrix} = \begin{pmatrix} 2.5 & 1.5 \\ 1.5 & 2.5 \end{pmatrix}$$

- Variance in X (2.5) and Y (2.5) is equal.
- Covariance (1.5) is positive, indicating a positive correlation between X and Y.

PCA Step 3: Eigen-Decomposition

Finding the Principal Components

- Now we treat the **Covariance Matrix (C)** like the matrix **A** from Part I.
- We find its Eigenvalues (λ) and Eigenvectors ($\mathbf{v}$) using $\det(\mathbf{C} - \lambda\mathbf{I}) = 0$.
- **Eigenvalues Result (Calculated):**

$$\lambda_1 = 4.0 \quad \text{and} \quad \lambda_2 = 1.0$$

- **Interpretation:**
 - The Eigenvalues represent the **variance** along the direction of their corresponding Eigenvectors.
 - $\lambda_1 = 4.0$ is the largest variance.

PCA Step 4: Determining Principal Components

The New Basis Vectors

- The **Eigenvectors** of the Covariance Matrix are the **Principal Components** (PCs).
- They are the new axes that are orthogonal to each other.
- **Eigenvectors Result (Normalized):**

$$\mathbf{v}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \approx \begin{pmatrix} 0.707 \\ 0.707 \end{pmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \approx \begin{pmatrix} 0.707 \\ -0.707 \end{pmatrix}$$

- **PC1 ($\mathbf{v}_1$):** Corresponds to λ_1 . This is the direction of **maximum variance**.
- **PC2 ($\mathbf{v}_2$):** Corresponds to λ_2 . This is the direction of next maximum variance (perpendicular to PC1).

Variance Explained and Selection

Explained Variance

- **Total Variance:** Sum of all Eigenvalues $\Sigma\lambda = 4.0 + 1.0 = 5.0$.
 - **PC1 Proportion:** $\frac{\lambda_1}{\Sigma\lambda} = \frac{4.0}{5.0} = 0.8$ (**80%**).
 - **PC2 Proportion:** $\frac{\lambda_2}{\Sigma\lambda} = \frac{1.0}{5.0} = 0.2$ (**20%**).
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- **Selection (k):** To reduce dimensionality, we select the top k components. If we want to capture 80% of the information, we only need $k = 1$ (PC1).

PCA Step 5: Data Projection

Creating the Final Dataset

- 1 **Feature Vector ($\mathbf{W}$):** Create a matrix of the top k Principal Component vectors (Eigenvectors). For $k = 1$:

$$\mathbf{W} = \begin{pmatrix} 0.707 \\ 0.707 \end{pmatrix}$$

- 2 **Projection Formula:** Multiply the original centered data ($\mathbf{X}$) by the Feature Vector ($\mathbf{W}$).

$$\mathbf{X}_{projected} = \mathbf{X}\mathbf{W}$$

- 3 **Result:** A new dataset with only one dimension ($k = 1$), capturing 80% of the original dataset's variance.

Conclusion and Key Takeaways

Summary

- ① **Eigen-Decomposition** is the mathematical engine behind PCA.

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

- ② **Eigenvectors** define the special directions (Principal Components).
- ③ **Eigenvalues** quantify the variance along those directions.
- ④ **PCA** works by rotating the coordinate system to align with the data's axes of maximum variance, then discarding the axes with low variance.